

RETAIL MULTI-AGENT AI SYSTEM FOR FORECASTING & ANOMALY DETECTION

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PROBLEM STATEMENT

Traditional analytics systems often struggle to:

- Predict future demand accurately
- Detect abnormal retail behaviors
- Handle demand volatility
- Prevent stock shortages and overstock situations

These limitations negatively impact:

- Inventory management
- Operational efficiency
- Strategic business decision-making



PROJECT OBJECTIVES

Accurate Sales Forecasting:

Predict future demand patterns to optimize inventory planning.

Anomaly Detection:

Identify unusual patterns in retail time-series data.

Actionable Recommendations:

Generate data-driven business insights for strategic decisions.

Multi-Agent Orchestration:

Coordinate intelligent agents for seamless system operation.



GLOBAL ARCHITECTURE

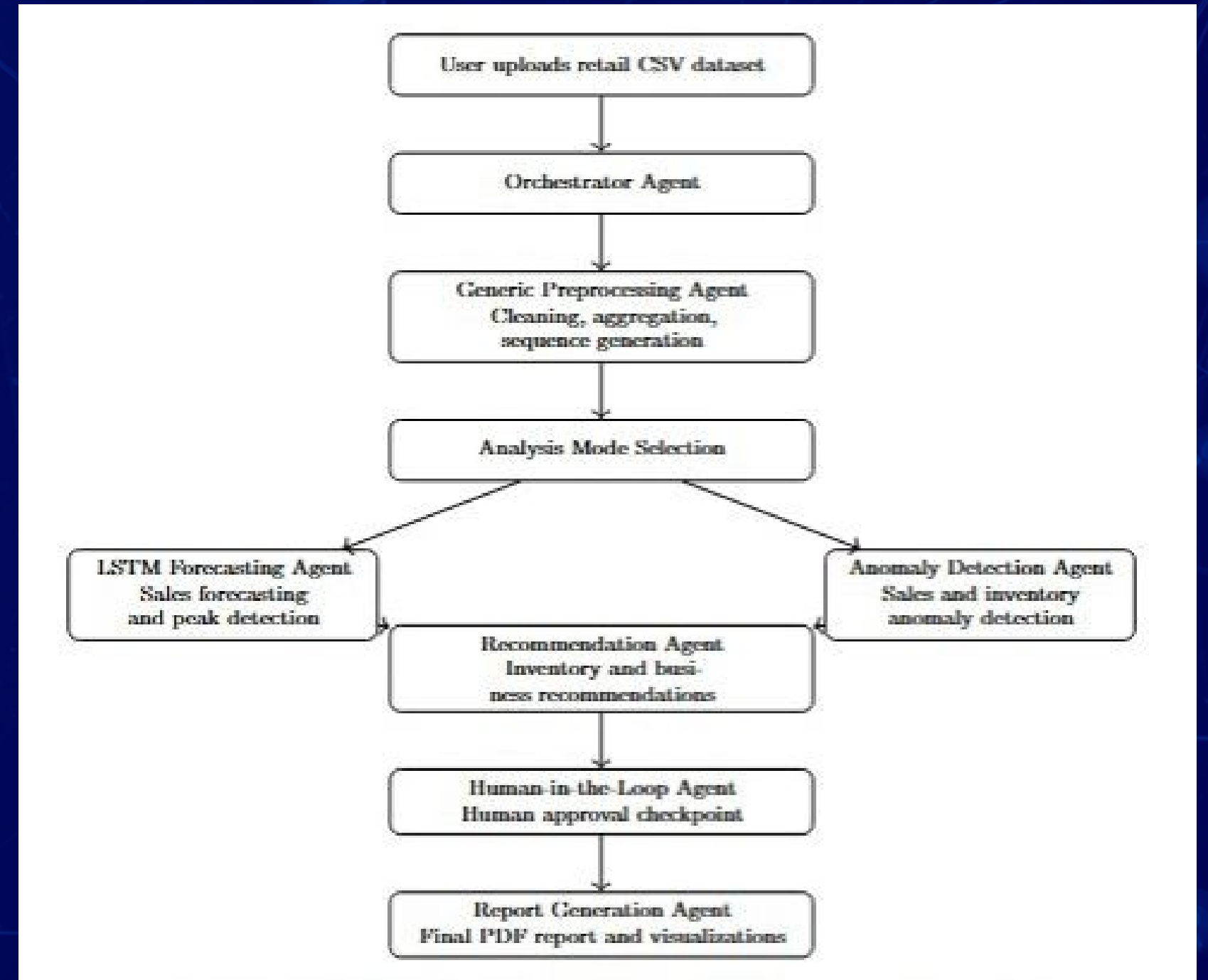
The proposed platform is based on a collaborative multi-agent architecture coordinated through a centralized orchestration workflow.

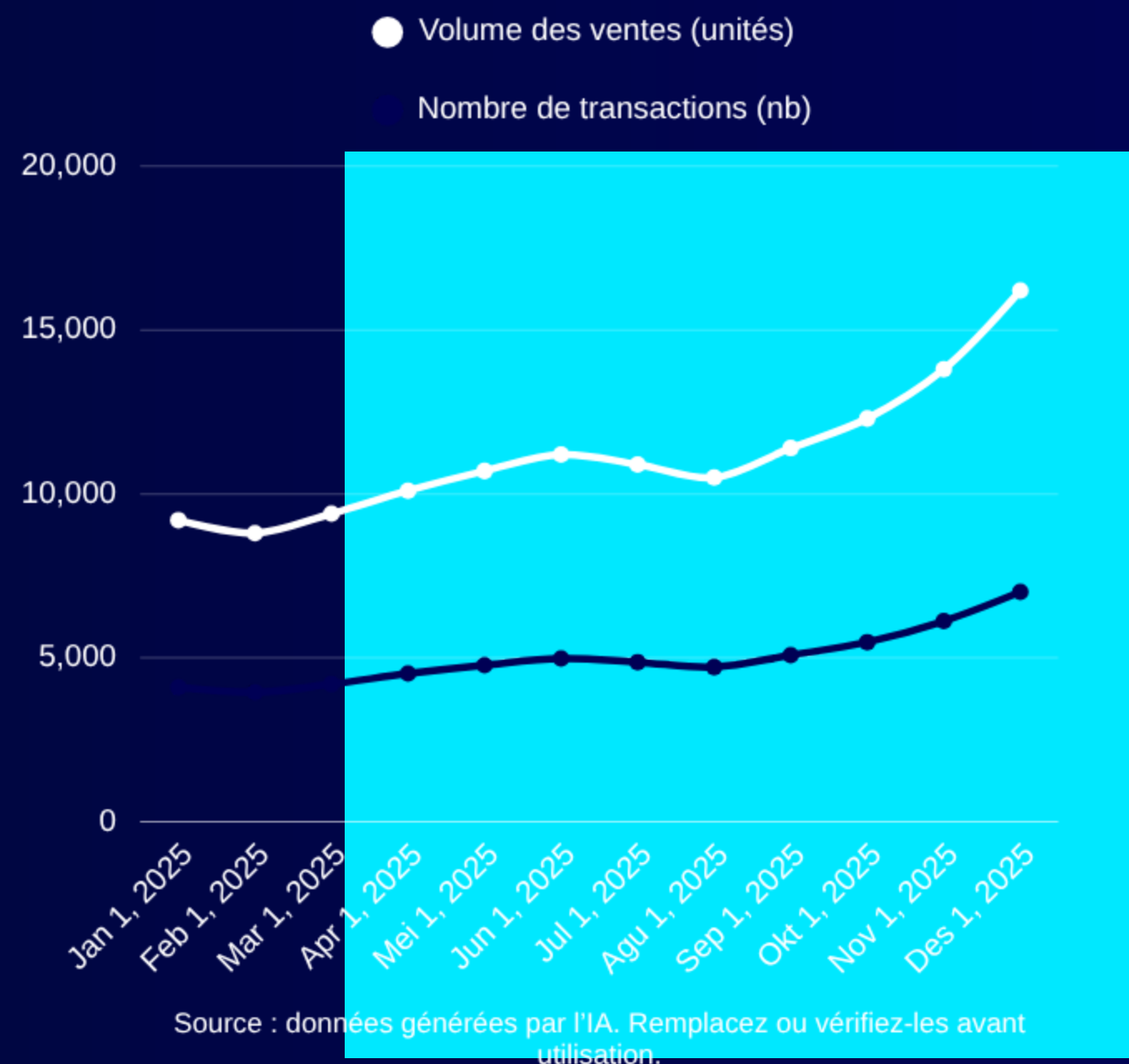
Main Components:

- Orchestrator Agent
- Preprocessing Agent
- Forecasting Agent
- Anomaly Detection Agent
- Recommendation Agent
- Human-in-the-Loop Agent
- Report Generation Agent

The architecture enables:

- Intelligent workflow coordination
- Modular AI analysis
- Automated retail analytics
- Business decision support





DATASET DESCRIPTION

- Retail time-series dataset
- Daily sales volumes
- Transaction count analysis
- Promotional event data
- External factors integration
- Oil price impact analysis
- Complex demand pattern modeling
- Forecasting and anomaly detection support

PREPROCESSING AGENT

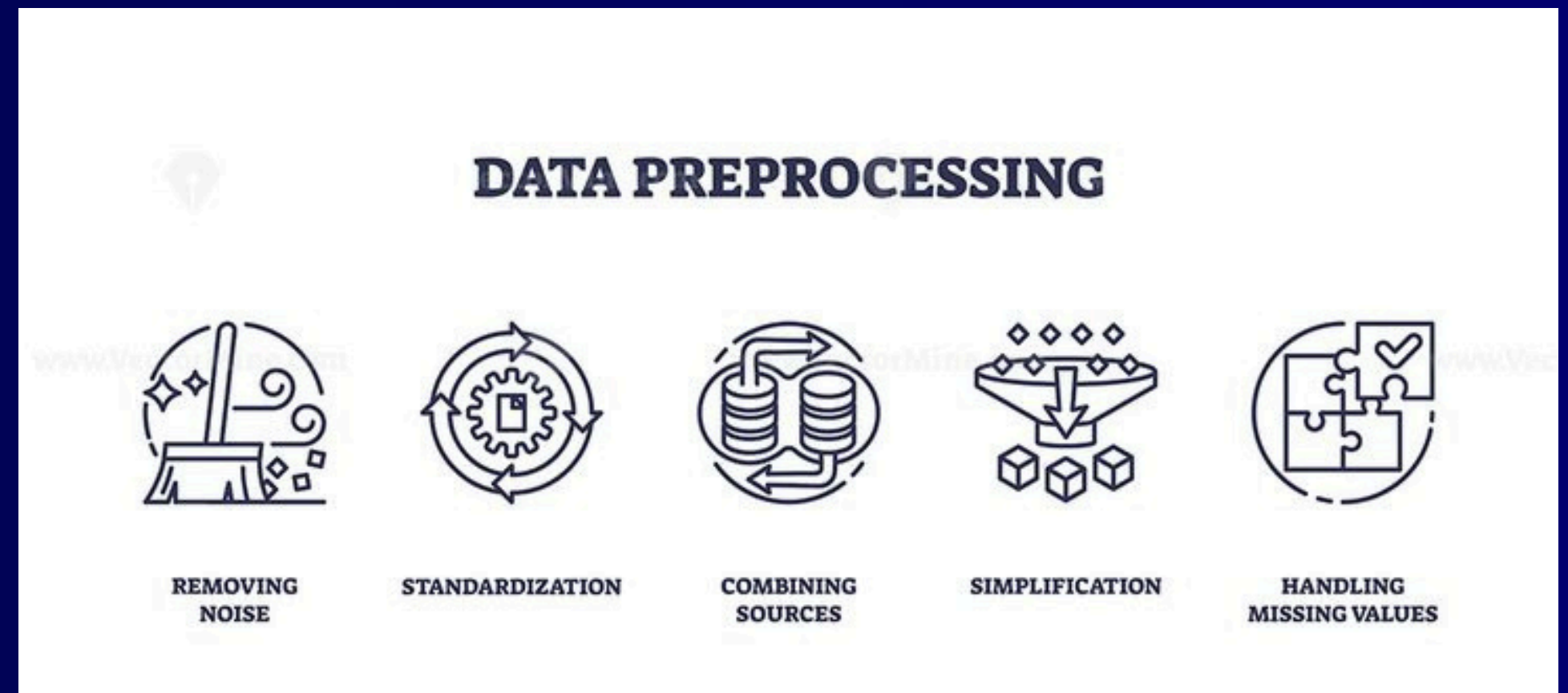
The Preprocessing Agent prepares retail datasets before model execution.

Main Responsibilities:

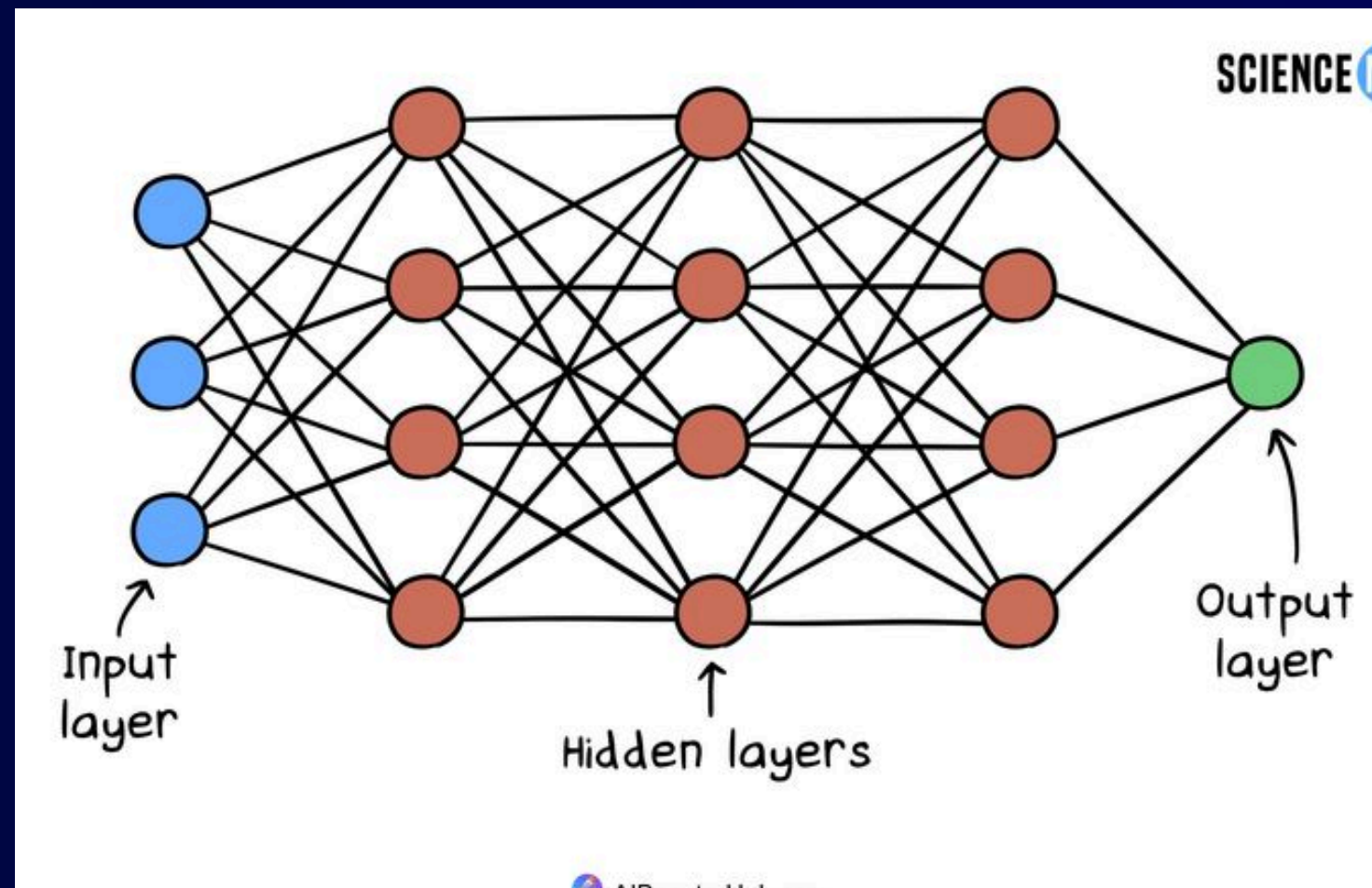
- CSV dataset loading
- Date and sales column detection
- Invalid data filtering
- Missing value handling
- Time-based feature generation
- Sequence preparation for LSTM models

Output:

A clean and model-ready retail time-series dataset.



FORECASTING AGENT



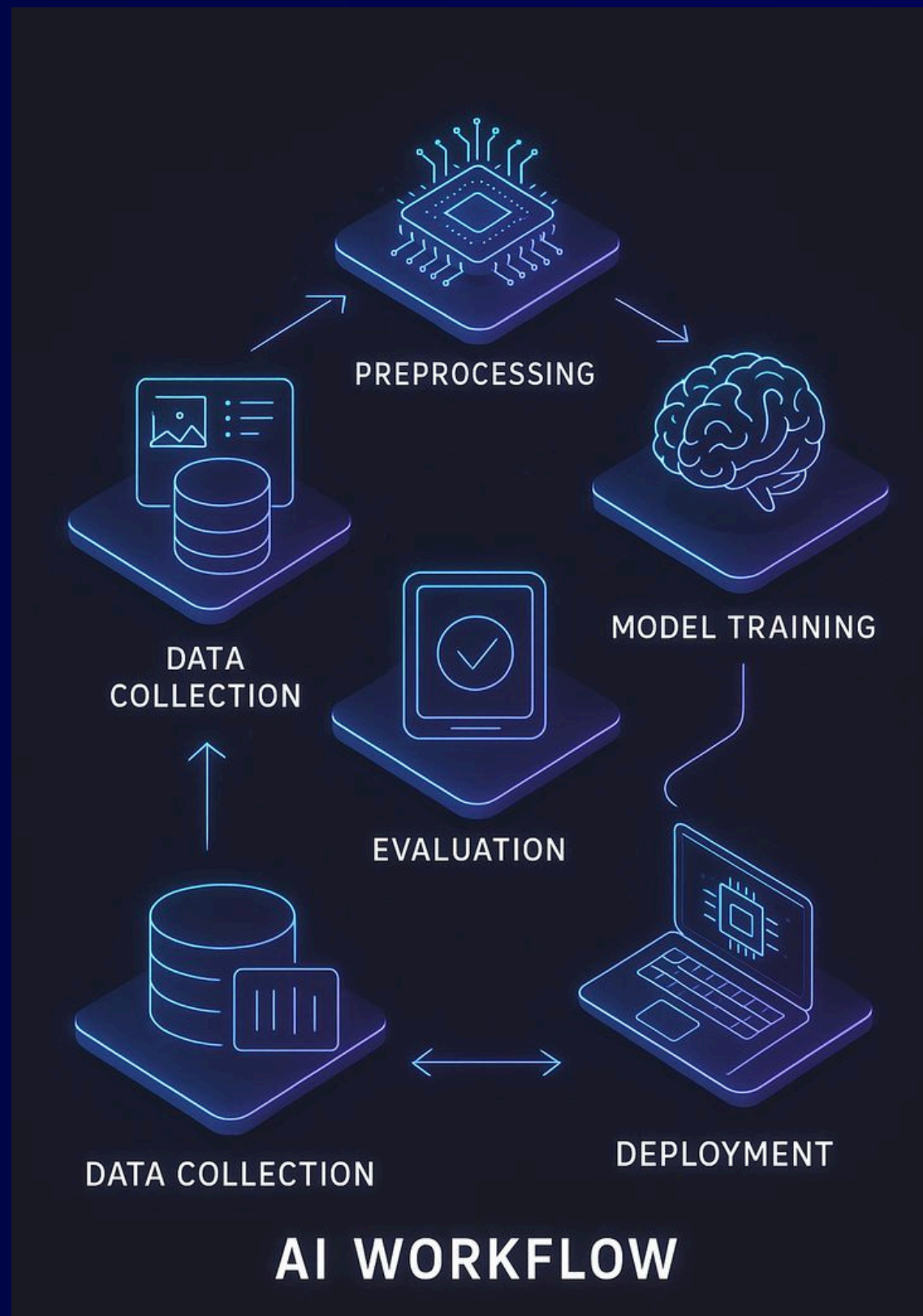
The Forecasting Agent uses a PyTorch LSTM model for retail demand prediction.

Input features include:

- historical sales
- lag values
- moving averages
- temporal features

Main functionalities:

- future sales forecasting
- peak demand detection
- trend analysis
- inventory estimation



WHY LSTM?

LSTM networks are highly effective for retail time-series forecasting because they can learn long-term temporal dependencies.

Advantages of LSTM:

- Sequential pattern learning
- Memory cell architecture
- Temporal dependency modeling
- Stable forecasting performance
- Long-range demand prediction

This makes LSTM well-suited for retail demand forecasting applications.



ANOMALY DETECTION AGENT

The anomaly detection module uses an LSTM Autoencoder to identify abnormal retail behaviors.

Detection is based on:

- reconstruction error
- sales spikes
- sales drops
- z-score analysis
- suspicious promotion patterns

Each anomaly is assigned a business risk level for operational interpretation.

RECOMMENDATION AGENT

The Recommendation Agent transforms forecasting and anomaly outputs into operational business recommendations.

Generated recommendations include:

- Inventory optimization
- Demand planning strategies
- Promotion adaptation
- Risk mitigation actions
- Operational decision support

The system provides actionable retail intelligence insights.



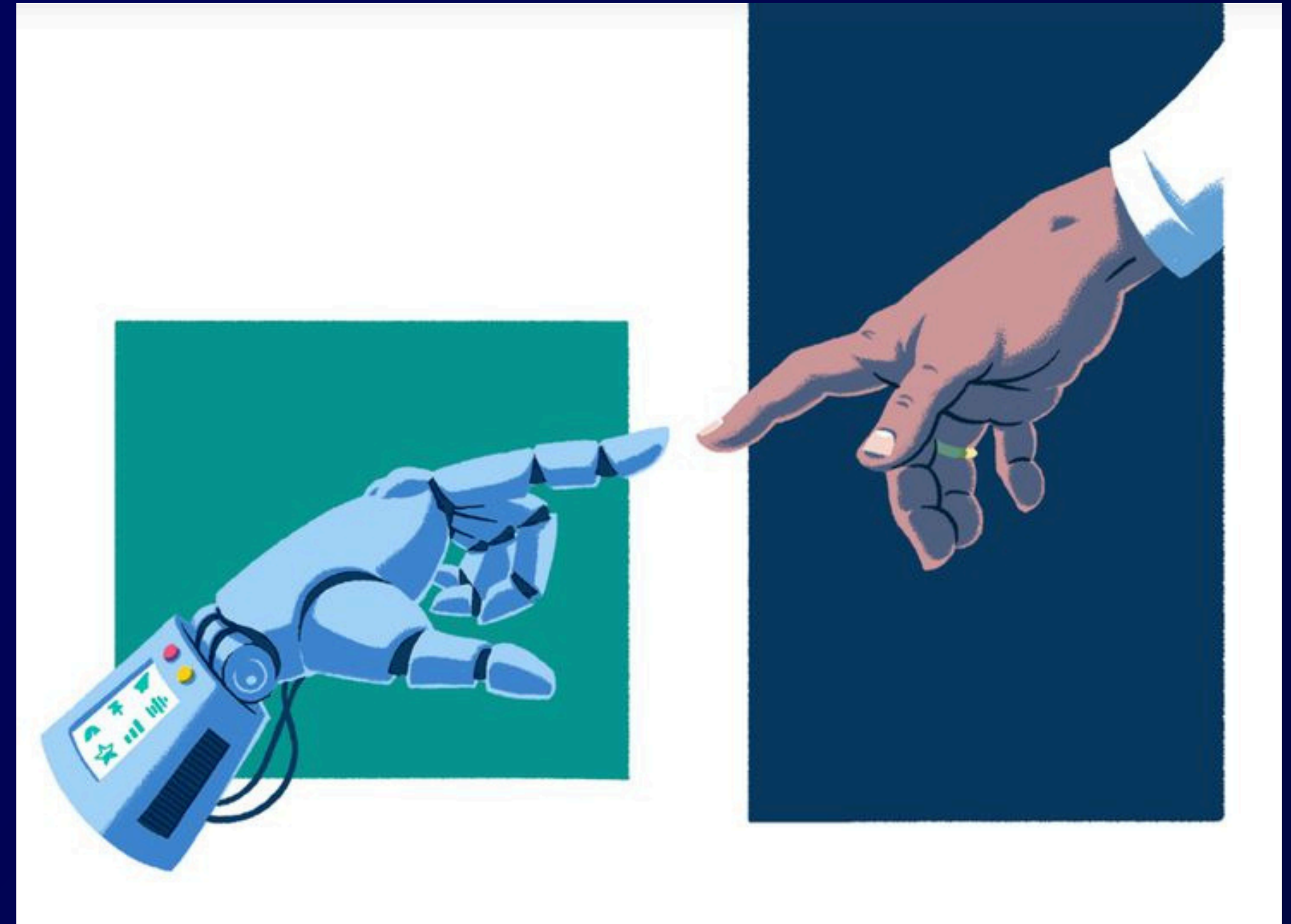
HUMAN-IN-THE-LOOP VALIDATION

Human validation is integrated to improve reliability and trustworthiness in AI-assisted business decisions.

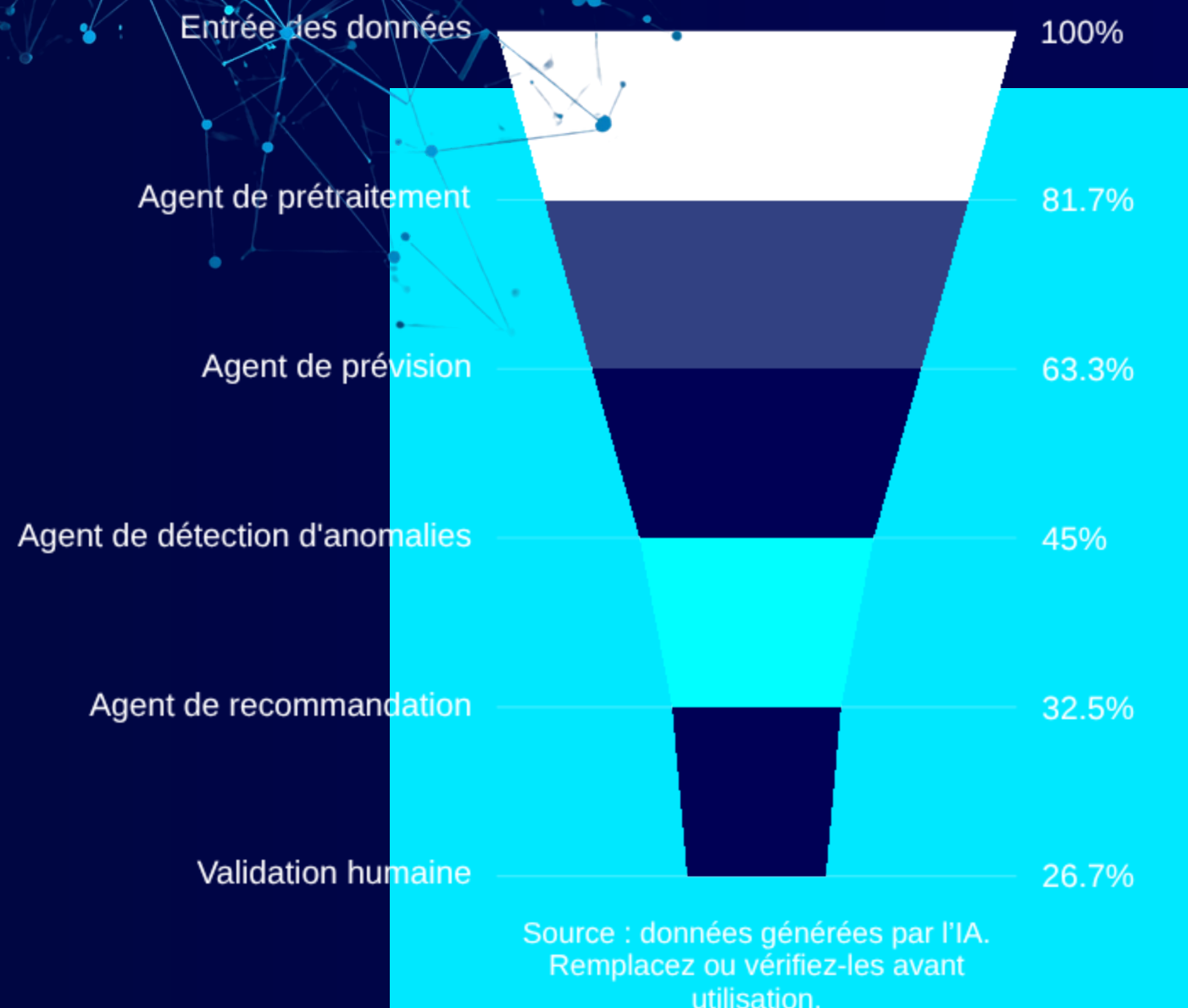
The Human-in-the-Loop Agent:

- Reviews recommendations
- Validates critical decisions
- Adds approval checkpoints
- Reduces operational risks

This approach ensures that important business actions are not fully automated.



Retail AI System



ORCHESTRATOR AGENT

- The Orchestrator Agent coordinates the complete multi-agent workflow.
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- Responsibilities:
 - Pipeline execution management
 - Agent communication
 - Task synchronization
 - Workflow monitoring
 - Error handling
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- The orchestrator ensures stable and automated execution of the retail analytics pipeline.

STREAMLIT DASHBOARD

The platform provides an interactive Streamlit dashboard for retail analytics visualization.

Dashboard functionalities:

- Retail dataset upload
- Analysis mode selection
- Forecast visualization
- Anomaly monitoring
- Metrics dashboards
- Automated report generation

The interface simplifies interaction with the AI platform.



TECHNØLOGIES USED

Python & PyTorch:

Core model development and deep learning implementation.

Streamlit:

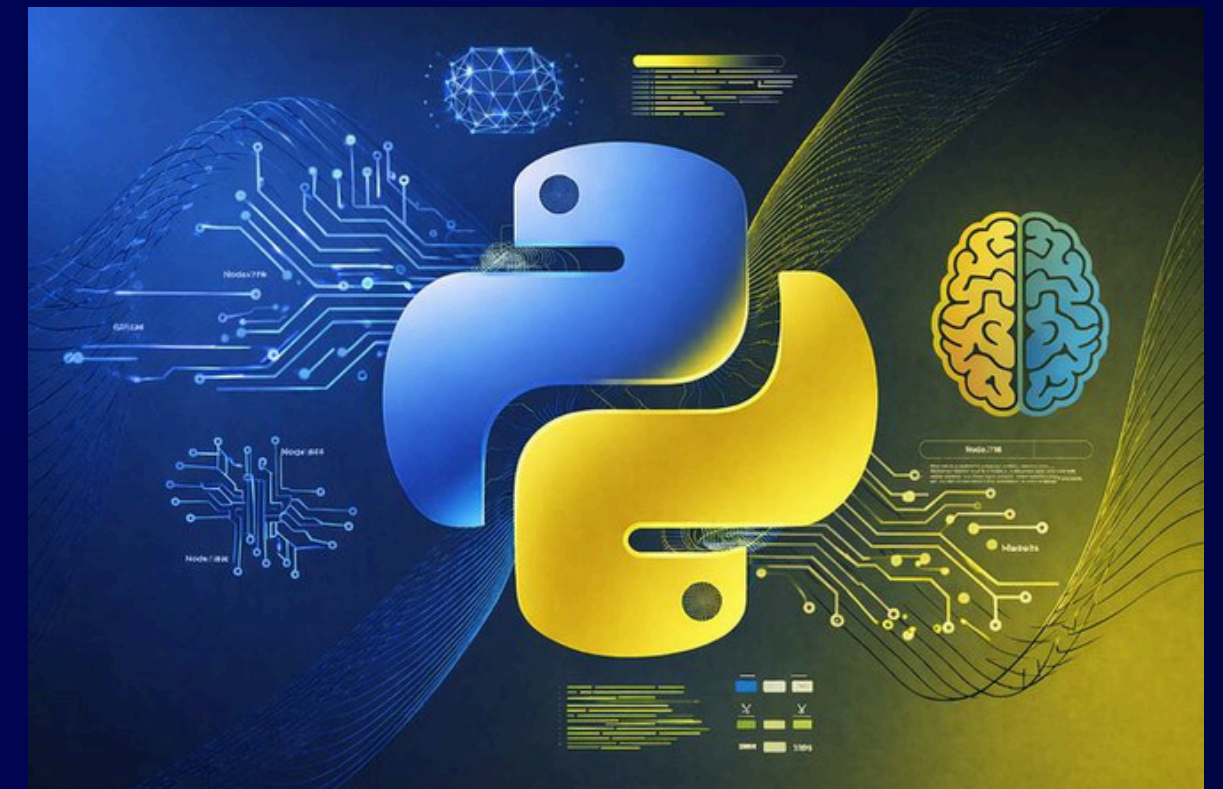
Interactive dashboard interface for visualization.

Pandas & Scikit-learn:

Data processing, feature engineering, and preprocessing pipelines.

Multi-Agent Framework:

Orchestration and agent coordination architecture.



RESULTS

FRetail AI System Results

Forecasting Metrics:

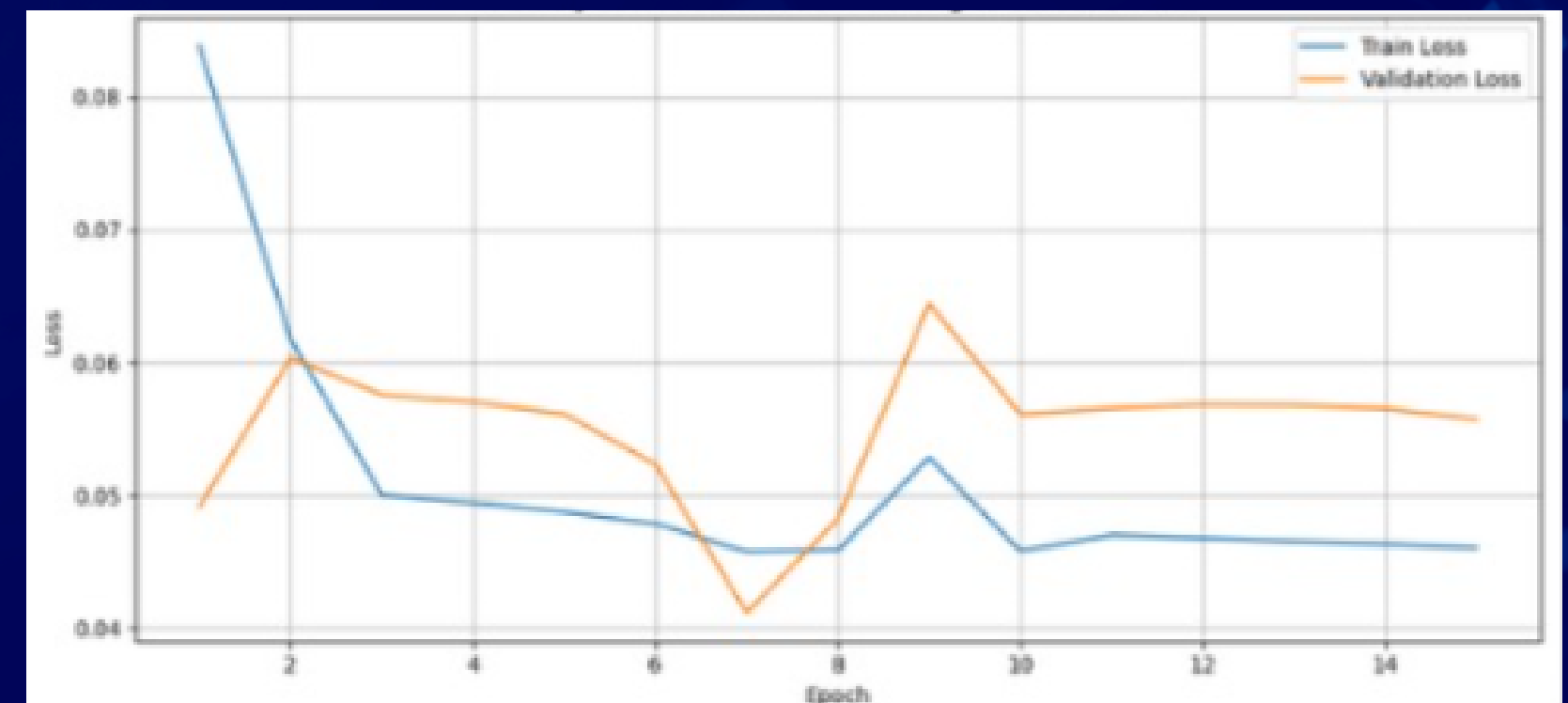
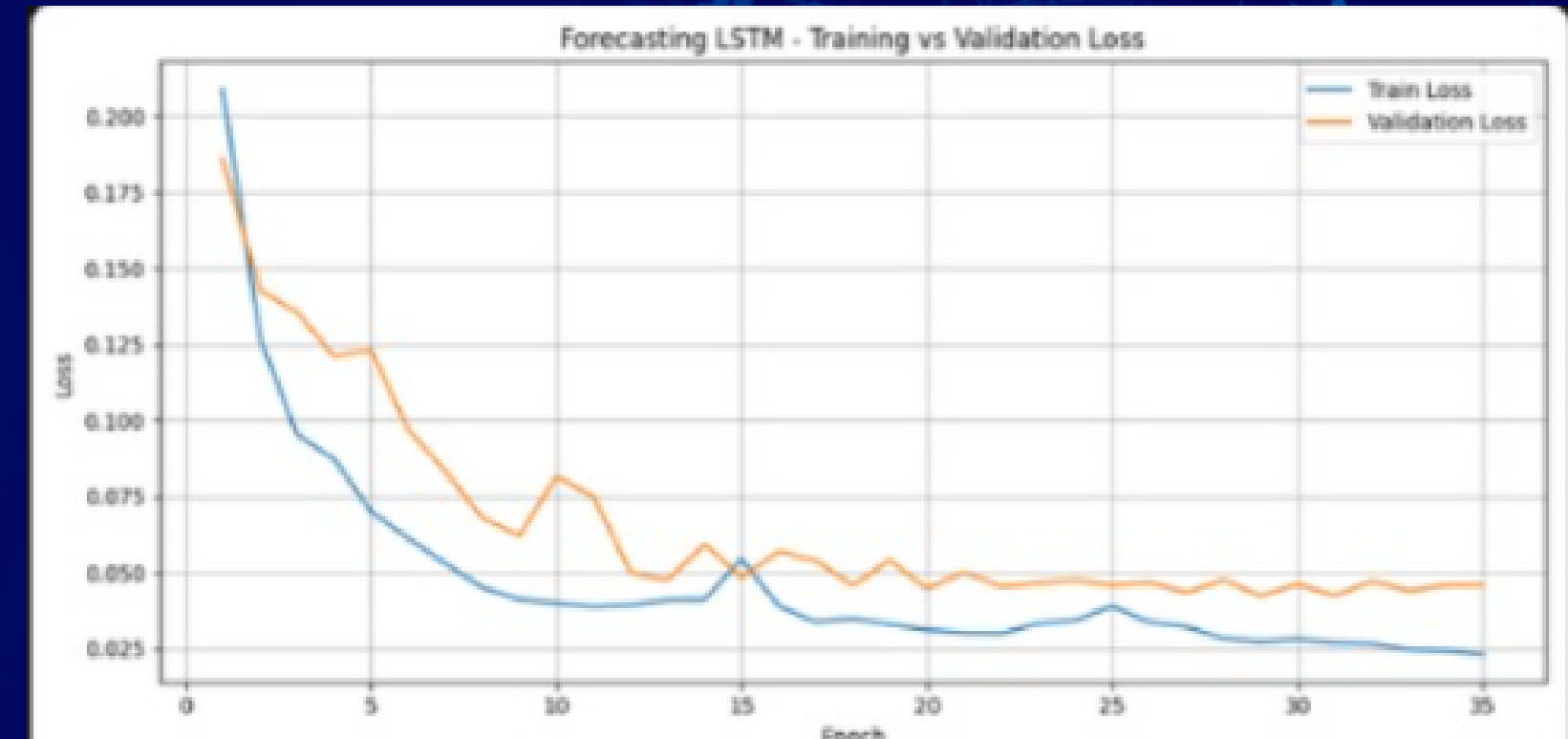
- MAE → 74,223
- RMSE → 121,430
- R² Score → 0.56
- SMAPE → 8.63%

Anomaly Detection:

- Reconstruction Loss → 0.046
- Anomaly Rate → 5.06%
- Detected Anomalies → 17

The forecasting model generated stable future retail demand predictions.

The anomaly detection model successfully identified abnormal retail behaviors and inconsistent demand patterns.



CHALLENGES & SOLUTIONS

Challenges

- Large retail time-series dataset processing
- Forecasting instability during training
 - Long Deep Learning training time
 - Slow model convergence
 - Temporal feature complexity
- Multi-agent workflow synchronization

Solutions

- Robust preprocessing pipeline
- Advanced feature engineering
- Hyperparameter optimization
 - Gradient clipping
 - Learning rate scheduling
- GPU acceleration with PyTorch
- Modular orchestration architecture

CONCLUSION

AI-driven retail intelligence enables smarter forecasting and demand planning. Deep learning techniques significantly improve anomaly detection accuracy and reliability. Multi-agent architecture facilitates a modular, robust, and scalable system design. This approach delivers significant business value and operational impact for retail operations.



THANK YOU

Any questions?